DATASCI 101 - Introduction to AI Applications

Final Project Instructions

Fall 2026

Overview

In groups of 4-5, you will study how AI is used in one real-world domain. First, you will test 2-3 existing AI tools in that domain and record what they do well and where they fail. Then you will use what you found to design a new AI application for a specific problem: what it does, how it works and what could go wrong.

No programming is required. We grade how clearly you think about AI systems: what they do, how they fail and how to build them responsibly.

Choosing a domain

Pick one domain from the table below. The examples are starting points; you can focus on any specific problem within your domain.

Domain	Example use cases
Healthcare	Symptom checking, mental health chatbots, medical image interpretation
Education	AI tutoring, automated grading, personalised learning platforms
Finance	Credit scoring, fraud detection, robo-advisors
Law	Contract review, legal research assistants, case-outcome prediction
Creative industries	AI-generated art, music or writing; content moderation
Scientific research	Literature search and summarisation, protein-structure prediction, experiment design
Customer service	Support chatbots, ticket routing and triage, sentiment analysis of feedback

Name your domain and use case when you submit your group list on 3 November. Domains are first come, first served. Several groups can share a domain, but each group must address a different use case. If a domain gets crowded, we may ask later groups to pick another. Email us if you are unsure which domain or use case to choose.

Timeline

Date	What happens
27 October (lecture 17)	Project instructions released
3 November	Group list due by email, with your domain and use case
5 November	Students without a group are assigned to one at random
12 November	Optional proposal due on Canvas
12-19 November	We send feedback on proposals
8 December	Report and infographic due on Canvas by 11:59 PM

Group formation

Form a group of 4-5 students by 3 November. One member emails the group list, domain and use case to `danilo.freire@emory.edu`. If you are not in a group by then, we will assign you to one at random on 5 November.

We especially recommend the optional proposal for randomly assigned groups. It gets you early feedback and a reason to start working together.

Deliverables

Proposal (optional, not graded)

Due 12 November on Canvas. Submit one page with three short paragraphs that answer these questions:

1. Which domain and use case did you choose, and why does the problem matter?
2. Which 2-3 AI tools will you test, and why these?
3. What application do you plan to design? What would it do, and who would use it?

We will send you written feedback, so you can change course before you write the full report.

Final submission (graded)

Due 8 December by 11:59 PM on Canvas. You submit two files: a report and an infographic.

Report

Write 8-12 pages of text and submit them as a PDF or Word document. Screenshots, figures, references and appendices do not count towards the limit. Each screenshot needs a sentence explaining what it shows and why it matters.

The report has five sections.

1. Introduction Describe the problem and explain why AI is relevant to it. A reader who does not know your domain should be able to follow.

2. Landscape review Test 2-3 existing AI tools in your domain. For each tool:

- Describe what it does
- Show how you tested it, with screenshots or sample prompts and outputs
- Discuss what it does well and where it falls short

Be specific. "The chatbot gave a wrong answer when asked about X" tells us more than "The chatbot sometimes makes mistakes".

3. Proposed application Use what you learned from testing to design your own AI application. Describe:

- What it would do

- Who would use it, and in what context
- What data it would need
- How it would work. No code is needed; a simple diagram or a description of the workflow is enough

4. Harms assessment Identify at least three things that could go wrong with your application, such as bias, privacy, reliability or misuse. For each risk, explain how you would prevent or reduce it. Use what you learned in class about AI ethics and safety.

5. Conclusion Summarise your main findings. What surprised you? What would you do differently with more time?

Infographic

Make a one-page PDF that summarises the problem, your proposed application and its main risks. Use any tool you like, such as Canva, PowerPoint or Google Slides. Someone who has not read your report should understand it.

Appendix (recommended, not graded)

Add an appendix with the material behind your report: the prompts you used, the full outputs, extra screenshots and any extended analysis. It does not count towards the page limit, and it lets us check the evidence for your landscape review.

Bonus: build a prototype (optional)

Build a simple working prototype with no-code tools. For example:

- A custom GPT, Claude Project or Gemini Gem with instructions for your use case
- A basic chatbot on a no-code platform such as Poe or Coze
- A simple retrieval-augmented generation (RAG) workflow in a visual tool
- A prompt chain that solves a multi-step task in your domain

Document it in a separate appendix, with screenshots and a short explanation of how it works. It does not count towards the page limit.

A working prototype with clear documentation earns 1 bonus point. A partial attempt, or one with little documentation, earns 0.5.

Grading rubric

The project is graded out of 10, plus up to 1 bonus point.

Component	What we look for	Points
Introduction	A clear problem statement that explains why AI matters here	1
Landscape review	Thorough testing of 2-3 tools; specific observations backed by screenshots or examples; a fair comparison	2
Proposed application	A realistic design; clear users, data needs and workflow; original thinking	3
Harms assessment	Risks specific to your application, not generic ones; concrete, realistic mitigations	2
Infographic	Clear, accurate and easy for a general audience to follow	1
Writing quality	Well organised; clear prose; proper citations; a conclusion that ties the report together	1
Total		**10**
Bonus: prototype	A working prototype with clear documentation (1 point) or a partial attempt (0.5 points)	**+1**

Tips

- Be specific. “This tool is biased” is too vague. “Given identical applications with different names, the credit-scoring tool scored the one with a typically white-sounding name 15% higher” is much better
- Use the lectures on bias, documentation, hallucination and harms assessment. All four apply directly to this project
- Divide the work early. Six weeks is plenty of time for a group that starts early, and not enough if everyone waits until the last week
- Write the proposal. It takes about an hour and can save you from going down the wrong path
- Design the infographic with care. A good one shows that you understand your project

AI use policy

You may use AI tools such as ChatGPT, Claude or Gemini to brainstorm, draft or edit. These rules apply:

- Declare all AI use in your submission
- You are responsible for the accuracy of everything in your report
- Do not paste AI output without editing and checking it
- The Emory Honor Code applies

Questions?

If anything is unclear, email danilo.freire@emory.edu or come to office hours.